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Integral Representation of Convex Functionals on a Space of Measures

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In duality pairs such as $(\mathscr{M}^b, \mathscr{C}_0)$ and $(W^{-1,p'}, W_0^{1,p})$, a convex integral functional on the space of functions has a polar which admits an integral representation. This representation is the sum of a first term involving the absolutely continuous component of the measure and of a second one which is a positively homogeneous function of the singular part. The duality is useful in plasticity theory. In the Sobolev case the study of non-parametric integrands is new. A description of the sub-differential is obtained. © 1988 Academic Press, Inc.

INTRODUCTION

Our motivations arise from two kinds of problems.

FIRST PROBLEM. In the mathematical theory of plasticity the energy can be expressed by

$$\int_\Omega f(x, Du(x))\, dx,$$

where $f(x, \cdot)$ is convex with linear growth. The function u can be discontinuous so its gradient (more precisely its deformation) Du has to be taken in the distribution sense. With some appropriate hypotheses (see [36]), Du

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belongs to the space $\mathscr{M}^{\mathrm{b}}$ of bounded measures, hence the idea of extending the functional

$$I_f : v \mapsto \int_\Omega f(x, v(x))\, dx$$

from L^1 to $\mathscr{M}^{\mathrm{b}}$ by taking the $\sigma(\mathscr{M}^{\mathrm{b}}, \mathscr{C}_0)$ lower semi-continuous hull

$$\bar{F} : \lambda \mapsto \varliminf_{v \to \lambda} I_f(v).$$

Let us point out that the $\sigma(\mathscr{M}^{\mathrm{b}}, \mathscr{C}_0)$ topology is the one which provides relative compactness of the sequence Du_ε when u_ε approaches the equilibrium.

When I_f is convex and proper one has

$$\bar{F}(\lambda) = \sup\{\langle \lambda, \varphi \rangle - I_{f^*}(\varphi) \mid \varphi \in \mathscr{C}_0\}.$$

The problem is to give an integral expression of $\bar{F}(\lambda)$.

SECOND PROBLEM. In the variational approach of semi-linear elliptic equations involving measures such as the Thomas–Fermi problem (see Brezis [13, 14] and Attouch, Bouchitté, and Mabrouk [2]), the Euler equation is obtained by computing the sub-differential on the Sobolev space $W_0^{1,p}$ of an integral functional $\int j(x, u(x))\, dx$. Usually the domain of the polar functional is contained in $\mathscr{M}^{\mathrm{b}} \cap W^{-1,p'}$.

Thus the two problems lead to the calculus on a space of measures of the polar of an integral functional. When f or j do not depend on x, the expression of the polar is due to Temam [37] and Demengel and Temam [19] for the first problem (but already in Valadier [40, 41]), and Brezis [11] completed by Grun–Rehomme [23] for the second one.

In the two previous problems it is important to allow f and j to depend on x (non-homogeneous media in the first situation and second member measure in the second one). In this direction the duality $(\mathscr{M}^{\mathrm{b}}, \mathscr{C}_0)$ has been considered by several authors (Rockafellar [32], Olech [28, 29], Valadier [41]). In the same way Giaquinta, Modica, and Soucek [21] and Dal Maso [16], using a result of Reschetniak [30], obtain the integral representation of $\bar{F}$ under hypotheses implying the continuity of f in (x, z) and its linear growth in z. Since 1985 this problem has been intensively studied by Hadhri [24], Valadier [42] (using Tran cao Nguyen [38, 39]), and De Giorgi, Ambrosio, and Buttazzo [17].

Our approach is new. It reduces the calculus of

$$\sup\left\{\int \varphi \cdot d\lambda - \int f^*(\cdot, \varphi)\, d\mu \mid \varphi \in \mathscr{C}_0\right\}$$

to the calculus of

$$[J+\delta(\cdot\,|\,\mathscr{C}_0)]^*\left(\frac{d\lambda}{dm}\right),$$

where m is a positive measure such that $\mu \ll m$ and $\lambda \ll m$, and J $(=I_{f^*})$ is an integral functional with respect to m. The basic result (Theorem 1 of Section 2) may seem rather abstract but it contains almost all difficulties. On the whole the proof is shorter than those of all previous paper.

In Section 3 we recover the formula (already in Valadier [40])

$$\bar{F}(\lambda)=\int g\left(\cdot,\frac{d\lambda_a}{d\mu}\right)d\mu+\int h\left(\cdot,\frac{d\lambda_s}{d\,|\lambda_s|}\,d\,|\lambda_s|\right),$$

where $\lambda_a+\lambda_s$ is the Lebesgue decomposition (with respect to μ) of λ and the integrands h and g derive from f. The situation is quite different from the non-parametric case where $g=f$ and $h=f_\infty$ the recession function of f. Indeed as shown in the examples of Section 5, g can be different from f. Nevertheless, under some regularity assumptions which are set in Section 4, the equality $h=f_\infty(x,\cdot)$ may occur μ-a.e. (which implies $g(x,\cdot)=f(x,\cdot)$ a.e.) or everywhere. A comparison is then possible with the results of [1, 16, 21].

The application to the duality $(W_0^{1,p}, W^{-1,p'})$ (second problem) is studied in [5, 7, 8]; the results of Brezis [11] and Grun–Rehomme [23] are extended.

The present paper follows and improves in some details on Bouchitté [4, 5, 6]. Sections 2 to 4 include the results of Valadier [42], with new proofs, and some other results (especially in Section 4).

1. Notations

Throughout this paper Ω denotes a locally compact metrizable space which is σ-compact, that is, a union of a countable sequence of compact subsets. This allows Ω to be compact metrizable (which from the mathematical standpoint would be simpler). This also allows Ω to be an open subset of $\mathbb{R}^N$.

A positive Radon measure μ on Ω is given. When Ω is an open subset of $\mathbb{R}^N$ it may be the Lebesgue measure. We will denote by m an auxiliary positive measure.

The space of continous functions tending to 0 at infinity is denoted by $\mathscr{C}_0(\Omega)$, abbreviated as $\mathscr{C}_0$. The space of $\mathbb{R}^d$-valued functions $\mathscr{C}_0(\Omega;\mathbb{R}^d)$ is also denoted by $[\mathscr{C}_0]^d$ and d will often be omitted. By $\mathscr{C}_c$ we denote the space of continuous functions with compact supports. When Ω is an open

subset of $\mathbb{R}^N$, $\mathscr{C}^\infty$ is the space of infinitely differentiable functions and $\mathscr{C}_c^\infty$ or $\mathscr{D}$ is the subspace of functions with compact supports.

By $\mathscr{M}$ and $\mathscr{M}^b$ we denote respectively the spaces of Radon measures on Ω and of bounded measures. The spaces of $\mathbb{R}^d$-valued measures are denoted by $\mathscr{M}(\Omega; \mathbb{R}^d)$, $\mathscr{M}^b(\Omega; \mathbb{R}^d)$ or $[\mathscr{M}]^d$, $[\mathscr{M}^b]^d$ (d will often be omitted).

Most of the paper uses one of the duality pairs $(\mathscr{M}, \mathscr{C}_c)$ or $(\mathscr{M}^b, \mathscr{C}_0)$. The bilinear form is denoted with brackets (for example $\langle \lambda, \varphi \rangle$) but the scalar product of $z, z' \in \mathbb{R}^d$ is denoted by $z \cdot z'$. If F is a function on a vector space E, F^* denotes its polar

$$F^*(x') = \sup\{\langle x', x\rangle - F(x) \mid x \in E\}$$

and $\operatorname{dom} F = \{x \mid F(x) < \infty\}$. If C is a subset of E, $\delta(\cdot \mid C)$ denotes its indicator function (taking value 0 on C, $+\infty$ outside) and $\delta^*(\cdot \mid C)$ its support function.

A *normal integrand* f is a measurable function $f: \Omega \times \mathbb{R}^d \to \bar{\mathbb{R}}$. We say that f is a convex normal integrand if moreover, $\forall x$, $f(x, \cdot)$ is convex l.s.c.

Other notation: $\mathbb{N}$ is the set of integers $n \geqslant 0$, $\mathbb{N}^* = \mathbb{N} \setminus \{0\}$, $\bar{B}(x, r)$ is the closed ball with center x and radius r, and δ_a is the Dirac measure at a.

2. Preliminary Results

2.1. We denote by $\mathscr{L}^0(\Omega, m)$ the vector space of real measurable functions.

Definition. A subset $\mathscr{H}$ of $[\mathscr{L}^0]^d$ is said to be *PCU-stable* if for any continuous partition of unity $(\alpha_0, ..., \alpha_n)$ such that $\alpha_1, ..., \alpha_n$ belong to $\mathscr{C}_c$ (variant, when Ω is an open subset of $\mathbb{R}^N$, $\alpha_1, ..., \alpha_n \in \mathscr{D}(\Omega)$, $\alpha_0 \in \mathscr{C}^\infty(\Omega)$), for every $u_0, ..., u_n$ in $\mathscr{H}$, $\sum_{i=0}^n \alpha_i u_i$ belongs to $\mathscr{H}$.

Remark. In the main applications $\mathscr{H}$ will be $[\mathscr{C}_0]^d$ or $[\mathscr{C}_c]^d$ and, in other papers [5, 7, 8], $\{\tilde{u} \mid u \in [W_0^{1,p} \cap L^\infty]^d\}$, where $\tilde{u}$ denotes all quasi-continuous elements of the Lebesgue equivalence class of u ([3, 12]).

2.2. Recall the following result [43, Proposition 1.14] (for a more recent paper see Fougères [20]). For any subset $\mathscr{H}_1$ of $[\mathscr{L}^0]^d$ there exists a smallest closed-valued measurable multifunction Γ such that $\forall u \in \mathscr{H}_1$, $u(x) \in \Gamma(x)$ m-a.e. (smallest refers to inclusion a.e.). We write $\Gamma = \operatorname{ess\,sup}_{u \in \mathscr{H}_1}\{u(\cdot)\}$ and say that Γ is the *essential supremum* of the multifunctions $x \mapsto \{u(x)\}$ $(u \in \mathscr{H}_1)$. Moreover there exists a sequence (u_n) in $\mathscr{H}_1$ such that a.e. $\Gamma(x) = \operatorname{cl}\{u_n(x) \mid n \in \mathbb{N}\}$. If (v_n) is any other sequence in $\mathscr{H}_1$ we can add the v_n to the u_n. Thus if $\mathscr{H}_1 \subset [\mathscr{C}_0(\Omega)]^d$, since $\mathscr{C}_0$ is separable (for the uniform convergence norm), we can add a dense sequence and this

proves $\Gamma(x) = \mathrm{cl}\{u(x) \mid u \in \mathscr{H}_1\}$. If $\mathscr{H}_1$ is convex it is easy to see that Γ is (a.e.) convex valued. This remains true if $\mathscr{H}_1$ is PCU stable. Indeed for any compact subset K of Ω and $r_0, ..., r_n \geqslant 0$ such that $\sum r_i = 1$, there exists a continuous partition of unity $(\alpha_0, ..., \alpha_n)$ with $\alpha_1, ..., \alpha_n \in \mathscr{C}_c$ and $\forall i$, $\alpha_i(x) = r_i$ on K. Then adding to the u_n, all the $\sum \alpha_i u_i$ for $(\alpha_0, ..., \alpha_n)$ corresponding to rational r_i and K running through a countable family of compacts (K_p) such that $\bigcup K_p = \Omega$, one can easily check that $\Gamma(x)$ is convex.

2.3. Let $j: \Omega \times \mathbb{R}^d \to]-\infty, \infty]$ be a normal convex integrand. For any $u \in [\mathscr{L}^0]^d$, $j(\cdot, u)$ denotes the function $x \mapsto j(x, u(x))$. Denote J the functional

$$\left| \begin{array}{l} u \mapsto \displaystyle\int_\Omega j(\cdot, u)\, dm \\ [\mathscr{L}^0]^d \to \overline{\mathbb{R}}, \end{array} \right.$$

where, as usual in convex analysis, $\int j(\cdot, u)\, dm = +\infty$ as soon as $\int j(\cdot, u)^+\, dm = +\infty$.

THEOREM 1. *Let $\mathscr{H}$ be a PCU-stable subset of $[\mathscr{L}^0]^d$. Suppose $\exists u_0 \in \mathscr{H}$ with $J(u_0) \in \mathbb{R}$. Then $\Gamma = \operatorname{ess\,sup}_{u \in \mathscr{H} \cap \operatorname{dom} J}\{u(\cdot)\}$ is convex valued,*

$$\inf_{u \in \mathscr{H}} J(u) = \int_\Omega [\inf_{z \in \Gamma(x)} j(x, z)]\, m(dx)$$

and

$$\inf_{z \in \Gamma(x)} j(x, z) = \operatorname*{ess\,inf}_{u \in \mathscr{H} \cap \operatorname{dom} J} j(\cdot, u).$$

Commentary. Classical results about commutativity of $\int$ and inf assume that $\mathscr{H}$ is a decomposable vector space or the set of measurable selectors of a multifunction: see Rockafellar [31, 33], Hiai and Umegaki [25], and Bourass and Valadier [9].

Remark/Example. We cannot take $\Gamma = \operatorname{ess\,sup}_{u \in \mathscr{H}}\{u(\cdot)\}$. Indeed let $\Omega = \mathbb{R}$, m the Lebesgue measure, $d = 1$, K a compact subset of $\mathbb{R}$ such that $\mathrm{int}(K) = \varnothing$ and $m(K) > 0$ (one can construct K analogously to the Cantor set). Let

$$j(x, z) = \begin{cases} z & \text{if } x \in K \\ \delta(z \mid \{0\}) & \text{otherwise.} \end{cases}$$

Let $\mathscr{H} = \mathscr{C}_c$. Then $\inf_{u \in \mathscr{H}} J(u) = 0$ because, if $u \not\equiv 0$, the set $\{x \mid u(x) \neq 0$

and $x \notin K\}$ is open and non-empty, so has >0 measure and $J(u) = +\infty$. But $\operatorname{ess\,sup}_{u \in \mathscr{C}_c}\{u(\cdot)\}$ is the constant multifunction $x \mapsto \mathbb{R}$ and

$$\inf_{z \in \mathbb{R}} j(x, z) = \begin{cases} -\infty & \text{if } x \in K \\ 0 & \text{otherwise.} \end{cases}$$

Proof. (1) First $\mathscr{H} \cap \operatorname{dom} J$ is still PCU-stable (because $j(\cdot, \sum \alpha_i u_i) \leqslant \sum \alpha_i j(\cdot, u_i)^+$), hence Γ is convex valued.

(2) Prove the first equality.

Let $\gamma(x) = \inf_{z \in \Gamma(x)} j(x, z)$ (γ is μ-measurable; Castaing and Valadier [15, Lemma III.39]). First $\geqslant$ holds because, $\forall u \in \mathscr{H} \cap \operatorname{dom} J$, $u(x) \in \Gamma(x)$ a.e. so

$$j(x, u(x)) \geqslant \gamma(x) \qquad \text{a.e.}$$

Prove now $\leqslant$. Let $r \in \mathbb{R}$, $r > \int \gamma \, dm$. Thanks to Bourbaki [10] or Dellacherie and Meyer [18, Théorème 48, pp. 107–108] there exists α l.s.c. integrable such that $\forall x$, $\alpha(x) \geqslant \gamma(x)$ and $\int \alpha \, dm < r$ (as $\gamma^+ \leqslant j(\cdot, u_0)^+$, γ^+ is integrable and can be approached upper by a l.s.c. function, and γ^- can be approached below by an u.s.c. function). We may modify slightly α to obtain $\forall x$, $\alpha(x) > \gamma(x)$.

Let $(u_n)_{n \geqslant 1}$ be a sequence in $\mathscr{H} \cap \operatorname{dom} J$ such that $\Gamma(x) = \operatorname{cl}\{u_n(x) \mid n \in \mathbb{N}^*\}$. Let N be a negligible set such that $\forall n$, $\forall x \in \Omega \backslash N$, $j(x, u_n(x)) \in \mathbb{R}$ (recall that $u_n \in \operatorname{dom} J$ implies $j(\cdot, u_n)^+$ is integrable and that $j(x, z) > -\infty$). Let $\varepsilon > 0$. There exists K compact, $K \subset \Omega \backslash N$ such that $\int_{\Omega \backslash K} [|j(\cdot, u_0)| + |\alpha|] \, dm < \varepsilon$. There exists $\eta > 0$ such that $m(A) < \eta$ implies $\int_A [|j(\cdot, u_0)| + |\alpha|] \, dm < \varepsilon$. Let K^ε be a compact such that $K^\varepsilon \subset K$, $m(K \backslash K^\varepsilon) < \eta$ and $\forall n$, $j(\cdot, u_n)$ is continuous on K^ε.

Let $A_n = \{x \in K^\varepsilon \mid j(x, u_n(x)) < \alpha(x)\}$. It is an open subset of K^ε. From Lemma A1 (see Appendix 1) applied with $D = \{u_n(x) \mid n \in \mathbb{N}^*\}$ (so $\bar{D} = \Gamma(x)$), for any $x \in K^\varepsilon$, $\gamma(x) = \inf_{n \geqslant 1} j(x, u_n(x))$, hence $\bigcup_{n \geqslant 1} A_n = K^\varepsilon$. By compactness there exists p such that $K^\varepsilon = \bigcup_{n=1}^p A_n$. There exists an open subset V^ε of Ω such that $V^\varepsilon \supset K^\varepsilon$ and

$$\forall n, \qquad 0 \leqslant n \leqslant p \Rightarrow \int_{V^\varepsilon \backslash K^\varepsilon} j(\cdot, u_n)^+ \, dm < \frac{\varepsilon}{p+1}.$$

Let V_n be a relatively compact open subset of Ω such that $V_n \cap K^\varepsilon = A_n$. We may suppose $V_n \subset V^\varepsilon$. There exists a continuous partition of unity $(\alpha_0, ..., \alpha_p)$ such that $\forall i = 1, ..., p$, $\operatorname{supp} \alpha_i \subset V_i$ and $\operatorname{supp} \alpha_0 \subset \Omega \backslash K^\varepsilon$ (see, for example, Bourbaki [10, Chap. III.1, n° 2, Lemme 1, p. 43]; when Ω is an open subset of $\mathbb{R}^N$ it is possible to get $\forall i$, $\alpha_i \in \mathscr{C}^\infty(\Omega)$, see L. Schwartz [35, Chap. I, Théorème II]).

Let $u=\sum_{n=0}^{p}\alpha_n u_n$. As $\mathscr{H}$ is PCU-stable, $u\in\mathscr{H}$. One has

$$j(x,u(x))\leqslant\sum_{n=0}^{p}\alpha_n(x)\,j(x,u_n(x))\leqslant\begin{cases}\alpha(x) & \text{if}\quad x\in K^\varepsilon\\ \sum_{n=0}^{p} j(x,u_n(x))^+ & \text{if}\quad x\in V^\varepsilon\backslash K^\varepsilon\end{cases}$$

$$j(x,u(x))=j(x,u_0(x))\qquad\text{if}\quad x\in\Omega\backslash V^\varepsilon.$$

Then

$$\int_\Omega j(\cdot,u)\,dm\leqslant\int_{K^\varepsilon}\alpha\,dm+\int_{V^\varepsilon\backslash K^\varepsilon}\sum_0^p j(\cdot,u_n)^+\,dm+\int_{\Omega\backslash V^\varepsilon}|j(\cdot,u_0)|\,dm.$$

We have

$$\int_{K^\varepsilon}\alpha\,dm=\int_\Omega\alpha\,dm-\left(\int_{\Omega\backslash K}\alpha\,dm+\int_{K\backslash K^\varepsilon}\alpha\,dm\right)$$

$$\leqslant\int_\Omega\alpha\,dm+2\varepsilon\leqslant r+2\varepsilon$$

$$\int_{V^\varepsilon\backslash K^\varepsilon}\sum_0^p j(\cdot,u_n)^+\,dm\leqslant\varepsilon$$

$$\int_{\Omega\backslash V^\varepsilon}|j(\cdot,u_0)|\,dm\leqslant\int_{\Omega\backslash K^\varepsilon}\cdots$$

$$=\int_{\Omega\backslash K}\cdots+\int_{K\backslash K^\varepsilon}\cdots\leqslant 2\varepsilon.$$

Finally, $\int_\Omega j(\cdot,u)\,dm\leqslant r+5\varepsilon$.

(3) As shown in (2), $\gamma(x)=\inf_{n\geqslant 1}j(x,u_n(x))$ a.e. Hence $\gamma\geqslant\operatorname{ess\,inf}_{u\in\mathscr{H}\cap\operatorname{dom}J}j(\cdot,u)$. Conversely there exists a sequence (v_k) in $\mathscr{H}\cap\operatorname{dom}J$ such that

$$\operatorname*{ess\,inf}_{u\in\mathscr{H}\cap\operatorname{dom}J}j(\cdot,u)=\inf_k j(\cdot,v_k).$$

But $v_k(x)\in\Gamma(x)$ a.e. so

$$\gamma(x)\leqslant\inf_k j(x,v_k(x)).$$

THEOREM 2. *We keep the hypotheses of Theorem* 1. *Let $\mathscr{X}$ and $\mathscr{Y}$ be vector spaces of $\mathbb{R}^d$-valued measurable functions such that $\forall u\in\mathscr{X}$, $\forall v\in\mathscr{Y}$, $u(\cdot)\cdot v(\cdot)$ is m-integrable and $\mathscr{H}\subset\mathscr{X}$. Then, in the duality $(\mathscr{X},\mathscr{Y})$*

$$\forall v\in\mathscr{Y},\qquad [J+\delta(\cdot\mid\mathscr{H})]^*(v)=\int_\Omega k(\cdot,v)\,dm,$$

where $k(x,\cdot)=[j^*(x,\cdot)\nabla\delta^*(\cdot\mid\Gamma(x))]^{**}$ (here ∇ denotes the infimum convolution [27]).

Remark. It is possible with a minoration hypothesis to obtain that the $\sigma(\mathscr{X}, \mathscr{Y})$ l.s.c. hull of $J+\delta(\cdot \mid \mathscr{H})$ is $u \mapsto J(u)+\int_\Omega \delta(u(x) \mid \Gamma(x))\, m(dx)$ (see Bouchitté [5, Théorème 2]).

Proof.

$$\begin{aligned}[J+\delta(\cdot \mid \mathscr{H})]^*(v) &= \sup_{u \in \mathscr{X}} [\langle u, v\rangle - J(u) - \delta(u \mid \mathscr{H})] \\ &= \sup_{u \in \mathscr{H}} \int [u(\cdot)\cdot v(\cdot) - j(\cdot, u)]\, dm \\ &= -\inf_{u \in \mathscr{H}} \int j'(\cdot, u)\, dm\end{aligned}$$

with $j'(x, z) = j(x, z) - z \cdot v(x)$. Since dom $J' \cap \mathscr{X} = \operatorname{dom} J \cap \mathscr{X}$, the multifunction $\operatorname{ess\,sup}_{u \in \mathscr{H} \cap \operatorname{dom} J'} \{u(\cdot)\}$ is still Γ. Moreover $J'(u_0) \in \mathbb{R}$.

By Theorem 1,

$$\begin{aligned}[J+\delta(\cdot \mid \mathscr{H})]^*(v) &= -\int \inf_{z \in \Gamma(x)} [j(x, z) - z \cdot v(x)]\, m(dx) \\ &= \int [j(x, \cdot) + \delta(\cdot \mid \Gamma(x))]^*(v(x))\, m(dx).\end{aligned}$$

Since $j(x, \cdot)$ and $\delta(\cdot \mid \Gamma(x))$ are l.s.c.

$$j(x, \cdot) + \delta(\cdot \mid \Gamma(x)) = [j^*(x, \cdot)\, \nabla\delta^*(\cdot \mid \Gamma(x))]^*$$

(see, for example, Castaing and Valadier [15, Proposition I.19]).

It is possible to choose classical spaces for $\mathscr{X}$ and $\mathscr{Y}$.

PROPOSITION 3. *Let j be a normal convex integrand. Suppose $\mathscr{H}$ is a vector subspace of $[\mathscr{L}^\infty]^d$ such that $\forall u \in \mathscr{H}$, $\forall \alpha \in \mathscr{C}_c(\Omega)$ (variant, when Ω is an open subset of $\mathbb{R}^N$, $\forall \alpha \in \mathscr{D}(\Omega)$), αu belongs to $\mathscr{H}$. Suppose $\exists u_0 \in \mathscr{H}$ such that $J(u_0) \in \mathbb{R}$. Let $\Gamma = \operatorname{ess\,sup}_{u \in \mathscr{H} \cap \operatorname{dom} J} \{u(\cdot)\}$.*

(1) *Consider the functional on $[L^\infty]^d$, $J+\delta(\cdot \mid \mathscr{H})$. Then its polar on $[L^1]^d$ verifies*

$$[J+\delta(\cdot \mid \mathscr{H})]^*(v) = \int_\Omega k(\cdot, v)\, dm,$$

where $k(x, \cdot) = [j^*(x, \cdot)\, \nabla\delta^*(\cdot \mid \Gamma(x))]^{**}$.

(2) *If $\mathscr{H} \subset [\mathscr{C}_0]^d$ then $\Gamma(x) = \operatorname{cl}\{u(x) \mid u \in \mathscr{H} \cap \operatorname{dom} J\}$ a.e.*

Proof. Remark that $\mathscr{H}$ is PCU-stable because $\sum_{i=0}^n \alpha_i u_i = u_0 + \sum_{i=1}^n \alpha_i (u_i - u_0)$.

(1) This results from Theorem 2 applied with $\mathscr{X} = [\mathscr{L}^\infty]^d$ and $\mathscr{Y} = [\mathscr{L}^1]^d$.

(2) This has been said in 2.2.

Remark. It is possible to give a variant with $\mathscr{Y} = [\mathscr{L}^1_{\text{loc}}]^d$ and for $\mathscr{X}$ the space of $\mathscr{L}^\infty$-functions with compact supports.

3. Description of $\bar{F}$

Let $f: \Omega \times \mathbb{R}^d \to]-\infty, \infty]$ be a convex normal integrand. We suppose

(H1) $\exists \varphi_0 \in \mathscr{C}_c$, $\exists a \in L^1$ such that μ-a.e. in x, $\forall z$, $f(x, z) \geqslant \varphi_0(x) \cdot z - a(x)$ (equivalently $\exists \varphi_0 \in \mathscr{C}_c$ such that $I_{f^*}(\varphi_0) < \infty$).

(H2) $\exists u_0 \in [L^1_{\text{loc}}(\Omega, \mu)]^d$ such that $I_f(u_0) < \infty$ (equivalently $\exists u_0 \in [L^1_{\text{loc}}]^d$, $\exists b \in L^1$ such that μ-a.e., $\forall z$, $f^*(x, z) \geqslant z \cdot u_0(x) - b(x)$).

Here, for any $u \in [\mathscr{L}^0(\mu)]^d$, $I_f(u) = \int_\Omega f(\cdot, u)\, d\mu$. Let $F: [\mathscr{M}]^d \to]-\infty, \infty]$ be defined as

$$F(\lambda) = \begin{cases} I_f\left(\dfrac{d\lambda}{d\mu}\right) & \text{if } \lambda \ll \mu \\ +\infty & \text{otherwise.} \end{cases}$$

(Note that $d\lambda/d\mu \in L^1_{\text{loc}}$ and, by (H1), $f(\cdot, d\lambda/d\mu) \geqslant \varphi_0(\cdot) \cdot (d\lambda/d\mu)(\cdot) - a$, hence $F(\lambda) > -\infty$.)

THEOREM 4. *Let*

$$h(x, z) = \sup\{\varphi(x) \cdot z \mid \varphi \in \mathscr{C}_c \cap \operatorname{dom} I_{f^*}\}$$

$$g(x, \cdot) = [f(x, \cdot)\, \nabla h(x, \cdot)]^{**},$$

$\lambda \in [\mathscr{M}]^d$, $\lambda_a + \lambda_s$ its Lebesgue decomposition with respect to μ, θ any positive measure such that $\lambda_s \ll \theta$. Then the $\sigma(\mathscr{M}, \mathscr{C}_c)$ l.s.c. hull of F is

$$\bar{F}(\lambda) = \int_\Omega g\left(\cdot, \frac{d\lambda_a}{d\mu}\right) d\mu + \int_\Omega h\left(\cdot, \frac{d\lambda_s}{d\theta}\right) d\theta,$$

and the $\sigma(L^1_{\text{loc}}, \mathscr{C}_c)$ l.s.c. hull of I_f is I_g.

With

(H2)′ $\exists u_0 \in [L^1]^d$ such that $I_f(u_0) < \infty$, and $F_1: [\mathscr{M}^b]^d \to]-\infty, \infty]$ defined by

$$F_1(\lambda) = \begin{cases} I_f\left(\dfrac{d\lambda}{d\mu}\right) & \text{if } \lambda \ll \mu \\ +\infty & \text{otherwise} \end{cases}$$

we obtain

THEOREM 4'. *The $\sigma(\mathscr{M}^b, \mathscr{C}_0)$ l.s.c. hull $\bar{F}_1$ of F_1 is*

$$\bar{F}_1(\lambda) = \int_\Omega g\left(\cdot, \frac{d\lambda_a}{d\mu}\right) d\mu + \int_\Omega h\left(\cdot, \frac{d\lambda_s}{d\theta}\right) d\theta$$

with g and h defined as in Theorem 4. Moreover the $\sigma(L^1, \mathscr{C}_0)$ l.s.c. hull of I_f is I_g.

Remarks. (1) If (H1) were replaced by

(H1)' $\exists \varphi_0 \in \mathscr{C}_0$ such that $I_{f^*}(\varphi_0) < \infty$

one would have to redefine h and g.

(2) If μ is non-atomic one can start from a measurable integrand f not necessarily convex, and the l.s.c. hulls $\bar{F}$ and $\bar{F}_1$ are the same as those obtained starting from f^{**}; this results from the Liapunov theorem. See Valadier [41] and Bouchitté [5].

(3) As h is sublinear the choice of θ is immaterial as soon as $\lambda_s \ll \theta$. See Goffman and Serrin [22].

Proof of Theorem 4. First, since L^1_{loc} is decomposable and $I_f(u_0) < \infty$, thanks to a famous theorem by Rockafellar, the polar F^* of F in the duality $(\mathscr{M}, \mathscr{C}_c)$ is

$$F^*(\varphi) = \sup_{u \in L^1_{\text{loc}}} [\langle u, \varphi \rangle - I_f(u)] = I_{f^*}(\varphi).$$

Thanks to minoration (H1) and convexity, $\bar{F} = F^{**}$, hence

$$\bar{F}(\lambda) = \sup_{\varphi \in \mathscr{C}_c} [\langle \lambda, \varphi \rangle - I_{f^*}(\varphi)].$$

Consider now a fixed $\lambda \in [\mathscr{M}]^d$. There exists a Borel set A such that

$$\mu(\Omega \setminus A) = |\lambda_s|\,(A) = 0.$$

Let $m = \mu + |\lambda_s|$. Then $\lambda \ll m$ and

$$\frac{d\lambda}{dm}(x) = \begin{cases} \dfrac{d\lambda_a}{d\mu}(x) & \text{if } x \in A \\[2ex] \dfrac{d\lambda_s}{d|\lambda_s|}(x) & \text{if } x \in \Omega \setminus A. \end{cases}$$

Thus $d\lambda/dm \in L^1_{\text{loc}}(m)$. Setting

$$j(x, z) = \begin{cases} f^*(x, z) & \text{if } x \in A \\ 0 & \text{if } x \in \Omega \setminus A \end{cases}$$

one has

$$\langle \lambda, \varphi \rangle - \int_\Omega f^*(\cdot, \varphi)\, d\mu = \int_\Omega \frac{d\lambda}{dm} \cdot \varphi\, dm - \int_\Omega j(\cdot, \varphi)\, dm.$$

Now we can apply Theorem 2 with $\mathscr{X} = \mathscr{H} = \mathscr{C}_c$ and $\mathscr{Y} = [L^1_{\mathrm{loc}}]^d$. Indeed, by (H1) and (H2), $J(\varphi_0) \in \mathbb{R}$ (remark $J = I_{f^*}$). Thus

$$\begin{aligned}\bar{F}(\lambda) &= \sup_{\varphi \in \mathscr{C}_c} \int_\Omega \left[\frac{d\lambda}{dm} \cdot \varphi - j(\cdot, \varphi)\right] dm \\ &= [J + \delta(\cdot \mid \mathscr{C}_c)]^* \left(\frac{d\lambda}{dm}\right) \\ &= \int_\Omega k\left(\cdot, \frac{d\lambda}{dm}\right) dm\end{aligned}$$

with $k(x, \cdot) = [j^*(x, \cdot) \nabla \delta^*(\cdot \mid \Gamma(x))]^{**}$.

Since $\Gamma(x) = \mathrm{cl}\{\varphi(x) \mid \varphi \in \mathscr{C}_c \cap \mathrm{dom}\, I_{f^*}\}$ (in fact Γ is defined up to equality m-a.e. but this expression is independent of m),

$$\delta^*(z \mid \Gamma(x)) = h(x, z).$$

Since

$$j^*(x, z) = \begin{cases} f(x, z) & \text{if } x \in A \\ \delta(z \mid \{0\}) & \text{if } x \in \Omega \setminus A, \end{cases}$$

$$k(x, \cdot) = \begin{cases} g(x, \cdot) & \text{if } x \in A \\ h(x, \cdot) & \text{if } x \in \Omega \setminus A. \end{cases}$$

Finally,

$$\begin{aligned}\bar{F}(\lambda) &= \int_A g\left(\cdot, \frac{d\lambda}{dm}\right) dm + \int_{\Omega \setminus A} h\left(\cdot, \frac{d\lambda}{dm}\right) dm \\ &= \int_\Omega g\left(\cdot, \frac{d\lambda_a}{d\mu}\right) d\mu + \int_\Omega h\left(\cdot, \frac{d\lambda_s}{d|\lambda_s|}\right) d|\lambda_s|.\end{aligned}$$

Proof of Theorem 4′. We still have, for $\varphi \in \mathscr{C}_0$, $F_1^*(\varphi) = I_{f^*}(\varphi)$ and

$$\bar{F}_1(\lambda) = \sup_{\varphi \in \mathscr{C}_0} [\langle \lambda, \varphi \rangle - I_{f^*}(\varphi)].$$

For a given $\lambda \in [\mathscr{M}^b]^d$, let A, m, and j be as in the proof of Theorem 4. Here $d\lambda/dm \in L^1(m)$.

We apply Theorem 2 with $\mathscr{Y} = \mathscr{L}^1$, $\mathscr{H} = \mathscr{C}_0$, and $\mathscr{X} = \mathscr{C}_0$ (or $\mathscr{L}^\infty$) (we may also apply Proposition 3). We get $\bar{F}_1(\lambda) = \int_\Omega k(\cdot, d\lambda/dm)\, dm$. Here the only difference is that

$$\Gamma(x) = \mathrm{cl}\{\varphi(x) \mid \varphi \in \mathscr{C}_0 \cap \mathrm{dom}\, I_{f^*}\}.$$

A priori, using $\mathscr{C}_0$ in place of $\mathscr{C}_c$ should give a greater function h. But let $\varphi \in \mathscr{C}_0 \cap \operatorname{dom} I_{f^*}$. There exists $\beta_n \in \mathscr{C}_c$, $\beta_n \geqslant 0$, $\beta_n \nearrow \chi_\Omega$, then $\psi_n = \beta_n \varphi + (1-\beta_n)\varphi_0$ (where φ_0 satisfies (H1)) belongs to $\mathscr{C}_c \cap \operatorname{dom} I_{f^*}$. Hence, for any x, $\psi_n(x) \to \varphi(x)$ and the function $\delta^*(z \mid \Gamma(x))$ is the same h as in Theorem 4.

THEOREM 5. *Under* (H1), *with h and g defined in Theorem* 4 *one has, for any bounded positive Borel function ψ, $\forall \lambda \in [\mathscr{M}]^d$ (or $[\mathscr{M}^b]^d$),*

$$\int_\Omega \psi g\left(\cdot, \frac{d\lambda_a}{d\mu}\right) d\mu + \int_\Omega \psi h\left(\cdot, \frac{d\lambda_s}{d\theta}\right) d\theta$$
$$= \sup\left\{\int_\Omega \psi\varphi \cdot d\lambda - \int_\Omega \psi f^*(\cdot, \varphi)\, d\mu \mid \varphi \in \operatorname{dom} I_{f^*} \cap \mathscr{C}_c \text{ (resp. } \mathscr{C}_0)\right\}.$$

Moreover, if ψ is continuous, the supremum can be taken on the whole space $\mathscr{C}_c$ or $\mathscr{C}_0$.

Comment. Consider the measure $G(\lambda)$ with values in $]-\infty, \infty]$ defined by, $\forall B$ Borel set,

$$[G(\lambda)](B) = \int_B g\left(\cdot, \frac{d\lambda_a}{d\mu}\right) d\mu + \int_B h\left(\cdot, \frac{d\lambda_s}{d\theta}\right) d\theta.$$

The first member in the statement is $\int \psi\, dG(\lambda)$. When $G(\lambda)$ is a Radon measure (equivalently takes finite values on compact sets) it is characterized by the knowledge of the values $\int \psi\, dG(\lambda)$, ψ continuous. The formula has been given by Temam [36, 37], Demengel and Temam [19], Hadhri [24], and Valadier [42, 45]. The continuity of ψ is necessary to take the supremum on $\mathscr{C}_0$.

Proof. (a) Consider for a fixed λ, $\lambda' = \psi\lambda$ and $m = \psi\mu + \psi\, |\lambda_s|$. Then $\lambda' \ll m$ and, if A is a Borel set such that $\mu(\Omega \setminus A) = |\lambda_s|\,(A) = 0$, one has

$$\frac{d\lambda'}{dm}(x) = \begin{cases} \dfrac{d\lambda_a}{d\mu}(x) & \text{if } x \in A \\[2ex] \dfrac{d\lambda_s}{d\,|\lambda_s|}(x) & \text{if } x \in \Omega \setminus A, \end{cases}$$

and, since ψ is bounded, $d\lambda'/dm \in L^1_{\mathrm{loc}}(m)$ (resp. $L^1(m)$). Set also

$$j(x, z) = \begin{cases} f^*(x, z) & \text{if } x \in A \\ 0 & \text{otherwise.} \end{cases}$$

Then $\int_\Omega \psi f^*(\cdot, \varphi)\, d\mu = \int_\Omega j(\cdot, \varphi)\, dm$, which will be denoted by $J(\varphi)$. Thus the right-hand side of the formula of Theorem 5 equals

$$\sup_{\varphi \in \mathscr{H}} \left[\int_\Omega \frac{d\lambda'}{dm} \cdot \varphi\, dm - J(\varphi) \right],$$

where $\mathscr{H} = \mathscr{C}_c \cap \operatorname{dom} I_{f^*}$ (or $\mathscr{C}_0 \cap \operatorname{dom} I_{f^*}$). Since ψ is bounded one has $\mathscr{H} \subset \operatorname{dom} J$, hence $\mathscr{H} \cap \operatorname{dom} J = \mathscr{H}$ and $\varphi_0 \in \mathscr{H} \cap \operatorname{dom} J$. Moreover $\mathscr{H}$ is PCU-stable. We can apply Theorem 2 with $\mathscr{Y} = [L^1_{\mathrm{loc}}]^d$ (or $[L^1]^d$), $\mathscr{X} = \mathscr{C}_c$ or $\mathscr{C}_0$. Thus

$$\sup_{\varphi \in \mathscr{H}} \left[\int_\Omega \frac{d\lambda'}{dm} \cdot \varphi\, dm - J(\varphi) \right] = \int_\Omega k\left(\cdot, \frac{d\lambda'}{dm} \right) dm$$

with $k(x, \cdot) = [j^*(x, \cdot)\, \nabla\delta^*(\cdot \mid \Gamma(x))]^{**}$ and $\Gamma = \operatorname{ess\,sup}_{u \in \mathscr{H}} \{u(\cdot)\}$. Again $\Gamma(x) = \operatorname{cl}\{\varphi(x) \mid \varphi \in \mathscr{H}\}$ and one can end the proof as in Theorem 4.

(b) Suppose that the supremum is on the whole space $\mathscr{C}_c$ (or $\mathscr{C}_0$) and that ψ is continuous. Proceeding as in (a), but with $\mathscr{H} = \mathscr{C}_c$ or $\mathscr{C}_0$, the difficulty is to check that, denoting $\Gamma = \operatorname{ess\,sup}_{u \in \mathscr{H} \cap \operatorname{dom} J} \{u(\cdot)\} = \operatorname{cl}\{\varphi(x) \mid \varphi \in \mathscr{C}_c$ or $\mathscr{C}_0$ and $\int \psi f^*(\cdot, \varphi)\, d\mu < \infty\}$, one has $\psi(x)\, \delta^*(z \mid \Gamma(x)) = \psi(x)\, h(x, z)$. We may suppose $\psi(x) > 0$. There exists a compact neighborhood K of x such that $\inf_K \psi = \delta > 0$. The remainder is routine.

4. Some Properties of h and g

Throughout this section the duality pair is either $(\mathscr{M}, \mathscr{C}_c)$ or $(\mathscr{M}^b, \mathscr{C}_0)$. Hypotheses (H1) and (H2) are assumed, so

$$\begin{aligned} h(x, z) &= \sup\{z \cdot \varphi(x) \mid \varphi \in \mathscr{C}_c \cap \operatorname{dom} I_{f^*}\} \\ &= \sup\{z \cdot \varphi(x) \mid \varphi \in \mathscr{C}_0 \cap \operatorname{dom} I_{f^*}\} \end{aligned}$$

(see the proof of Theorem 4′).

We will sometimes use in place of (H1) the stronger

(H1)″ $\exists \lambda_0 \in]0, \infty[$, $\exists a \in L^1$ such that a.e., $\forall z$, $f(x, z) \geqslant \lambda_0 |z| - a(x)$.

(Remark that (H1)″ $\Rightarrow$ (H1) with $\varphi_0 = 0$.)

Recall that the recession or asymptotic function $f_\infty(x, \cdot)$ of the convex l.s.c. proper function $f(x, \cdot)$ satisfies

$$\forall z_0 \in \operatorname{dom} f(x, \cdot), \qquad f_\infty(x, z) = \lim_{r \to \infty} \frac{f(x, z_0 + rz)}{r}$$

and $f_\infty(x, z) = \delta^*(z \mid \operatorname{dom} f^*(x, \cdot))$ (Rockafellar [34, Theorem 8.5, p. 66, and Theorem 13.3, p. 116]).

PROPOSITION 6. *Let*

$$E(x)=\left\{z\in\mathbb{R}^d \mid \exists V \text{ open}, V\ni x, \exists\varphi \text{ continuous on } V \text{ such that } \varphi(x)=z \text{ and } \int_V f^*(\cdot,\varphi)\,d\mu<\infty\right\}$$

$$E_1(x)=\left\{z\in\mathbb{R}^d \mid \exists V \text{ open}, V\ni x \text{ such that } \int_V f^*(\cdot,z)\,d\mu<\infty\right\}.$$

Then

(1) $\forall(x,z),\ h(x,z)=\delta^*(z\mid E(x))$,

(2) *if* $x\in\Omega\backslash\operatorname{supp}\mu$, $E(x)=E_1(x)=\mathbb{R}^d$ *and* $h(x,\cdot)=\delta(\cdot\mid\{0\})$,

(3) *under* (H1)″, $\forall x,\ E_1(x)\subset E(x)\subset\overline{E_1(x)}$.

EXAMPLE. Without (H1)″, (3) may be false. Let $\Omega=]-\pi,\pi[$, μ the Lebesgue measure, $d=2$,

$$D_x=\{\lambda(\cos x,\sin x)\mid\lambda\in\mathbb{R}\},\qquad f(x,\cdot)=\delta(\cdot\mid D_x).$$

Then $f^*(x,\cdot)=\delta(\cdot\mid D_x^\perp)$ and $E_1(0)=\{(0,0)\}$, $E(0)=\{0\}\times\mathbb{R}$.

Proof. (1) This is proved in Valadier [42, Proposition 7, p. 22] and is known since Olech [28].

(2) If $x\notin\operatorname{supp}\mu$, $V=\Omega\backslash\operatorname{supp}\mu$ is an open neighborhood of x and $\int_V f^*(x,z)\,\mu(dx)=0$ for any z. So $E(x)=E_1(x)=\mathbb{R}^d$ and $h(x,\cdot)=\delta(\cdot\mid\{0\})$.

(3) The inclusion $E_1(x)\subset E(x)$ is obvious. Let $z\in E(x)$. Let V and φ corresponding to z. We may, changing V in a smaller neighborhood, suppose φ bounded. For any $\varepsilon>0$, let $V_\varepsilon=\{y\in V\mid|\varphi(y)-z|<\varepsilon\}$ and W_ε a compact neighborhood of x contained in V_ε. There exists $\theta_\varepsilon\colon V\to[0,1]$ continuous such that $\theta_\varepsilon(x)=1$ on W_ε and $\operatorname{supp}\theta_\varepsilon\subset V_\varepsilon$. Define

$$\varphi_\varepsilon=\theta_\varepsilon z+(1-\theta_\varepsilon)\varphi.$$

Then $\varphi_\varepsilon(x)=z$ and $\sup_{y\in V}|\varphi_\varepsilon(y)-\varphi(y)|\leqslant\varepsilon$.

By (H1)″ the functional I on $L^\infty(V,\mu)$, defined by $I(v)=\int_V f^*(\cdot,v)\,d\mu$, is bounded on a (norm) neighborhood of 0, so it is continuous on $\operatorname{int}(\operatorname{dom} I)$, which contains $[0,\varphi[$. Hence if $r\in[0,1[$

$$\lim_{\varepsilon\to0}\int_V f^*(\cdot,r\varphi_\varepsilon)\,d\mu=\int_V f^*(\cdot,r\varphi)\,d\mu.$$

So for ε sufficiently small, $f^*(\cdot,r\varphi_\varepsilon)\in L^1$, hence $\int_{\operatorname{int}(W_\varepsilon)}f^*(\cdot,rz)\,d\mu<\infty$ and $rz\in E_1(x)$. Finally, $z\in\overline{E_1(x)}$.

PROPOSITION 7. (1) *One has μ-a.e.*

$$g(x, \cdot) \leqslant f(x, \cdot)$$
$$h(x, \cdot) = g_\infty(x, \cdot) \leqslant f_\infty(x, \cdot),$$

(2) *I_f is $\sigma(L^1, \mathscr{C}_0)$ (resp. $\sigma(L^1_{\mathrm{loc}}, \mathscr{C}_c)$) l.s.c. iff μ-a.e. $h(x, \cdot) = f_\infty(x, \cdot)$ (equivalently $h(x, \cdot) \geqslant f_\infty(x, \cdot)$).*

(3) *If Ω' is an open subset of Ω and if $x \mapsto \operatorname{epi} f^*(x, \cdot)$ is l.s.c. on Ω', then $\forall x \in \Omega'$, $f_\infty(x, \cdot) \leqslant h(x, \cdot)$. As a consequence if $\mu(\Omega \setminus \Omega') = 0$, I_f is l.s.c.*

EXAMPLE. Let $\Omega = \mathbb{R}$, μ the Lebesgue measure, $d = 1$, K a compact subset of $\mathbb{R}$ with $\operatorname{int}(K) = \varnothing$ and $\mu(K) > 0$, and

$$f(x, z) = \begin{cases} |z| & \text{if } x \in K \\ 0 & \text{otherwise,} \end{cases}$$

Then $I_{f^*}(\varphi) = \delta(\varphi \mid \{0\})$, so $\bar{I}_f = 0 \neq I_f$.

Proof. Parts (1) and (2) have been proved in Valadier [41, 42]. For a somewhat more direct proof see Bouchitté [5, 7].

(3) Let $x_0 \in \Omega'$. If $z_0 \in \operatorname{dom} f^*(x_0, \cdot)$, by the Michael theorem [26] there exists a continuous selector (φ, ψ) of $x \mapsto \operatorname{epi} f^*(x, \cdot)$ such that $(\varphi(x_0), \psi(x_0)) = (z_0, f^*(x_0, z_0))$. Let K be a compact neighborhood of x contained in Ω'. Then

$$\int_{\operatorname{int} K} f^*(\cdot, \varphi)\, d\mu \leqslant \int_K \psi\, d\mu < \infty.$$

Hence $z_0 \in E(x_0)$. Therefore $f_\infty(x_0, \cdot) \leqslant h(x_0, \cdot)$. The last assertion follows from (2).

THEOREM 8. (1) *Under one of the hypotheses*

(H3) *$\forall z$, $f^*(\cdot, z)$ is u.c.s. on Ω,*

(H4) *f is l.s.c. on $\Omega \times \mathbb{R}^d$ and $f(\cdot, 0)$ is locally bounded,*

one has $\forall x \in \Omega$, $f_\infty(x, \cdot) \leqslant h(x, \cdot)$ (hence I_f is l.s.c.).

(2) *Under* (H3) *or* (H4) *and moreover*

(H5) *$\forall z$, $f_\infty(\cdot, z)$ is u.c.s.,*

one has

$$h(x, z) = \begin{cases} f_\infty(x, z) & \text{if } x \in \operatorname{supp} \mu \\ \delta(z \mid \{0\}) & \text{if } x \in \Omega \setminus \operatorname{supp} \mu. \end{cases}$$

Remarks and Comments. (1) For I_f being $\sigma(L^1, \mathscr{C}_0)$ l.s.c. it is sufficient to have (H3) or (H4) on an open set Ω' such that $\mu(\Omega \setminus \Omega') = 0$, for example (as said in [24]) if

$$f(x, z) = \begin{cases} f_1(z) & \text{if} \quad x \in \Omega_1 \\ f_2(z) & \text{if} \quad x \in \Omega_2, \end{cases}$$

where Ω_1 and Ω_2 are disjoint open subsets such that

$$\mu(\Omega \setminus (\Omega_1 \cup \Omega_2)) = 0.$$

(2) If $f(x, \cdot)$ does not depend on x, (H3) and (H5) are obviously satisfied. If moreover $\operatorname{supp} \mu = \Omega$, the formula of Theorem 5 becomes, $\forall \psi$ Borel bounded positive function,

$$\sup \left\{ \int \psi \varphi \cdot d\lambda - \int \psi f^*(\cdot, \varphi)\, d\mu \mid \varphi \in \mathscr{C}_0 \cap \operatorname{dom} I_{f^*} \right\}$$
$$= \int \psi f\left(\frac{d\lambda_a}{d\mu}\right) d\mu + \int \psi f_\infty \left(\frac{d\lambda_s}{d|\lambda_s|}\right) d|\lambda_s|.$$

This is the starting formula (for ψ continuous) of Temam [37] and Demengel and Temam [19].

(3) In case f is l.s.c. on whole the space $\Omega \times \mathbb{R}^d$, hypothesis (H4), Giaquinta, Modica, and Soucek [21], and Dal Maso [16] obtain, thanks to a result of Reschetniak [30] about sublinear functions of measures, that the functional

$$G \left| \begin{array}{l} \lambda \mapsto \int f\left(\cdot, \dfrac{d\lambda_a}{d\mu}\right) d\mu + \int f_\infty \left(\cdot, \dfrac{d\lambda_s}{d|\lambda_s|}\right) d|\lambda_s| \\ [\mathscr{M}^{\mathrm{b}}]^d \to]-\infty, \infty] \end{array} \right.$$

is $\sigma(\mathscr{M}^{\mathrm{b}}, \mathscr{C}_0)$ l.s.c. As a consequence I_f is $\sigma(L^1, \mathscr{C}_0)$ l.s.c., hence $g(x, \cdot) = f(x, \cdot)$ μ-a.e. But it can happen that $G \neq \bar{F}_1$. Indeed consider the following example suggested in [16, 4.4, p. 414].

EXAMPLE. Let $\Omega = \mathbb{R}$, $\mu = dx$, $d = 1$.

$$f(x, z) = \begin{cases} |z| & \text{if} \quad |z|\,|x|^{1/2} \leqslant 1 \\ 2\,|z| - |x|^{-1/2} & \text{if} \quad |z|\,|x|^{1/2} \geqslant 1. \end{cases}$$

Then f is continuous on $\Omega \times \mathbb{R}$, (H4) is satisfied, but (H5) does not hold.

One can check that $\forall x$, $h(x, z) = 2\,|z|$ and $f_\infty(0, z) = |z|$. Thus $G(\delta_0) = 1$ and $\bar{F}_1(\delta_0) = 2$.

(4) In [1, 16] (where the more difficult problem of a functional depending on the gradient is studied), a sufficient condition ensuring $\bar{F}=G$ is set. This condition implies that f is continuous in x and has linear growth in z; more precisely,

$$\forall \varepsilon>0, \exists \delta>0, |x_1-x_2|<\delta \Rightarrow \forall z, |f(x_1, z)-f(x_2, x)| \leqslant \varepsilon(1+|z|).$$

This hypothesis is far more stringent that the one of (2) of Theorem 8. Indeed (H3) or (H4) supplemented with (H5) does not imply the continuity of $f(\cdot, z)$ but only the continuity of $f_\infty(\cdot, z)$ (remark that f being l.s.c., $f_\infty(\cdot, z)$ is l.s.c. too).

Proof of Theorem 8. (1) By Proposition 7 it is sufficient to prove that the multifunction $Q: x \mapsto \operatorname{epi} f^*(x, \cdot)$ is l.s.c.

(a) Under (H3). Let U be an open subset of $\mathbb{R}^d \times \mathbb{R}$. Then

$$\begin{aligned}\{x \in \Omega \mid Q(x) \cap U \neq \varnothing\} &= \{x \mid \exists (z, r) \in U \text{ such that } f^*(x, z) \leqslant r\} \\ &= \bigcup_{(z,r) \in U} \{x \mid f^*(x, z) < r\}\end{aligned}$$

(the change from $\leqslant$ to $<$ is easy) which is open.

(b) Under (H4). Recall that, for $(z, t) \in \mathbb{R}^d \times \mathbb{R}$,

$$\tilde{f}(x, z, t) = \delta^*((z, t) \mid Q(x)) = \begin{cases} -tf(x, z/-t) & \text{if } t<0 \\ f_\infty(x, z) & \text{if } t=0 \\ +\infty & \text{if } t>0. \end{cases}$$

From Lemma A2 it is sufficient to prove that $\tilde{f}$ is l.s.c. This is a consequence of Dal Maso [16].

(2) Under (H5)

$$\begin{aligned}V &= \{x \in \Omega \mid \exists z \in \mathbb{R}^d \text{ such that } f_\infty(x, z) < h(x, z)\} \\ &= \bigcup_z \{x \mid f_\infty(x, z) < h(x, z)\}\end{aligned}$$

is open (h defined in Theorem 4 is l.s.c.). From Proposition 7(1) a.e. $f_\infty(x, \cdot) \geqslant h(x, \cdot)$, so V is negligible, hence $V \cap \operatorname{supp} \mu = \varnothing$.

If $x \in \operatorname{supp} \mu$, $x \notin V$ and then using (1), $f_\infty(x, \cdot) = h(x, \cdot)$. If $x \notin \operatorname{supp} \mu$, the result follows from Proposition 6(2).

5. Examples

The proofs of the results stated in Examples 1 to 4 are left to the reader. For details see Bouchitté [5, 7].

EXAMPLE 1. Let Ω be an open subset of $\mathbb{R}^N$ and Σ be an $N-1$-dimensional hypersurface contained in Ω. Let μ denote the measure $dx + H^{N-1}(\Sigma \cap \cdot)$, where H^{N-1} is the $N-1$-dimensional Hausdorff measure (thus $H^{N-1}(\Sigma \cap \cdot)$ is the area measure of Σ). We suppose that Σ is regular, that is, μ is finite on compact sets and $\Omega \backslash \Sigma$ is dense in Ω.

Let

$$f(x, z) = \begin{cases} |z| & \text{if } x \in \Omega \backslash \Sigma \\ \frac{1}{2}|z|^2 & \text{if } x \in \Sigma, \end{cases}$$

Let

$$\beta(z) = \begin{cases} \frac{1}{2}|z|^2 & \text{if } |z| \leqslant 1 \\ |z| - \frac{1}{2} & \text{if } |z| \geqslant 1. \end{cases}$$

Remark that $\beta = \frac{1}{2}|\cdot|^2 \nabla |\cdot|$. Then, if $\lambda_a + \lambda_s$ is the μ-decomposition of λ,

$$\bar{F}(\lambda) = \int_{\Omega \backslash \Sigma} d\,|\lambda_a| + \int_{\Sigma} \beta\left(\frac{d\lambda_a}{d\mu}(x)\right) dH^{N-1}(x) + |\lambda_s|\,(\Omega).$$

EXAMPLE 2. Let Ω be an open subset of $\mathbb{R}^N$, μ the Lebesgue measure, $a: \Omega \to [0, \infty[$ a locally integrable function, and $f(x, z) = a(x)\,|z|$. Then, if

$$\tilde{a}(x) = \overline{\lim_{\delta \to 0_+}}\ [\mu(B(x, \delta))]^{-1} \int_{B(x,\delta)} a(y)\,dy$$

and $\hat{a}$ is the l.s.c. hull of $\tilde{a}$,

$$\bar{F}(\lambda) = \int_{\Omega} \hat{a}\,d\,|\lambda|.$$

Remark. As soon as $f^*(x, \cdot)$ is an indicator, $\bar{F}(\lambda) = \delta^*(\lambda \,|\, \Phi)$, where Φ is the set of $\mathscr{C}_c$-selectors of a l.s.c. multifunction Γ. For the existence of Γ see Valadier [44]. In Examples 2 and 4 below, it is possible to "calculate" Γ.

EXAMPLE 3. Let Ω be an open subset of $\mathbb{R}^N$, μ the Lebesgue measure, $a: \Omega \to [0, \infty[$ a measurable function, and $f(x, z) = \frac{1}{2}a(x)\,|z|^2$. Then, if Ω' is the greatest open subset on which $1/a$ is locally integrable (with the convention $1/0 = +\infty$), one has

$$\bar{F}(\lambda) = \begin{cases} \dfrac{1}{2}\displaystyle\int_{\Omega'} a(x) \left|\frac{d\lambda_a}{dx}\right|^2 dx & \text{if } |\lambda_s|\,(\Omega') = 0 \\ +\infty & \text{if } |\lambda_s|\,(\Omega') > 0. \end{cases}$$

EXAMPLE 4. Let Ω be an open subset of $\mathbb{R}^N$, μ the Lebesgue measure, and $A:\Omega \to \mathbb{R}^d$ a measurable function such that $|A(x)|=1$ a.e. Let $f(x,z)=[A(x)\cdot z]^+$. If $\tilde{A}$ is defined as in Example 2 but coordinate-wise, that is,

$$\forall i \in \{1, ..., d\}, \qquad \tilde{A}_i(x) = \overline{\lim_{\delta \to 0_+}} \, [\mu(B(x,\delta)]^{-1} \int_{B(x,\delta)} A_i(y)\, dy,$$

and if Ω' is the greatest open subset on which $\tilde{A}$ is continuous, then $\bar{F}(\lambda) = \int_{\Omega'} [(d\lambda/d\,|\lambda|)(x)\cdot \tilde{A}(x)]^+ \, |\lambda| \, (dx)$.

Remarks. (1) On Ω', $|\tilde{A}(x)|=1$ because $\tilde{A}(x)=A(x)$ a.e.

(2) The existence of Ω' and $\tilde{A}$ can be proved without the $\tilde{}$ operation. Indeed Ω' is the greatest open subset on which A is a.e. equal to a (unique) continuous function. The existence of Ω' follows from the Lindelöf property. One can treat also $f(x,z)=|A(x)\cdot z|$: in this case it is necessary to topologize the unit sphere identifying opposite points.

EXAMPLE 5 (which describes the usual case in plasticity theory). Let Ω be an open bounded subset of $\mathbb{R}^N$ and E the space of symmetric tensors of order 2 ($\dim E = N(N+1)/2$). Recall that E has a euclidean stucture for which the orthogonal of $\mathbb{R}I$ (the one-dimensional subspace of diagonal tensors) is the space E^D of rensors whose traces vanish.

Let B be a closed convex-valued l.s.c. multifunction such that $\forall x$, $0 \in B(x)$. We suppose moreover that $\forall \varphi \in \mathscr{C}_0$, $\varphi(x) \in B(x)$ a.e. $\Rightarrow \varphi(x) \in B(x)$ everywhere (remark that this avoids $\Omega =]-1, 1[$, $B(x)=[0,1]$ if $x \neq 0$, $B(0)=\{0\}$). There exist many l.s.c. discontinuous multifunctions which satisfy this hypothesis. In practice $B(x)=B^D(x)+\mathbb{R}I$, where $B^D(x)$ is a convex compact subset of E^D containing 0.

Let $\gamma: R \to \mathbb{R}$ be continuous and ψ be a convex normal integrand on $\Omega \times E$ such that $0 \leqslant \psi(x,\cdot) \leqslant \gamma(\cdot)$. The useful integrand in plasticity is

$$f(x,\cdot) = [\psi(x,\cdot) + \delta(\cdot \mid B(x))]^*.$$

Let $h(x,\cdot) = \delta^*(\cdot \mid B(x))$. Then

$$\forall \lambda \in \mathscr{M}^{\mathrm{b}}(\Omega; E), \qquad \bar{F}_1(\lambda) = \int_\Omega f\left(x, \frac{d\lambda_a}{dx}\right) dx + \int_\Omega h(x, d\lambda_s).$$

Remark. When $B(x)=B^D(x)+\mathbb{R}I$,

$$h(x,z) = \begin{cases} \delta^*(z \mid B^D(x)) & \text{if } z \in E^D \\ +\infty & \text{otherwise.} \end{cases}$$

Hence $\operatorname{dom} h(x, \cdot) = E^D$ and, if $u \in BD(\Omega)$ and $Du = \frac{1}{2}(u_{ij} + u_{ji})$ satisfies $\bar{F}_1(Du) < \infty$, the singular part of the measure $\operatorname{div} u = \operatorname{tr}(Du)$ vanishes.

Proof. Since $f^*(x, \cdot) = \psi(x, \cdot) + \delta(\cdot \mid B(x))$, one has for $\varphi \in \mathscr{C}_0$

$$I_{f^*}(\varphi) < \infty \Leftrightarrow \varphi(x) \in B(x) \text{ a.e.} \Leftrightarrow \forall x,\ \varphi(x) \in B(x).$$

Thanks to the Michael theorem [26], for any $z \in B(x)$, there exists $\varphi \in \mathscr{C}_0$ with $\varphi(x) = z$ and $\forall y,\ \varphi(y) \in B(y)$. Thus $h(x, \cdot) = \delta^*(\cdot \mid B(x))$ and, since $g^* = f^* + h^*$, $\forall x,\ g(x, \cdot) = f(x, \cdot)$.

Appendix 1

Lemma A1. *Let $g: \mathbb{R}^d \to]-\infty, \infty]$ be convex l.s.c., $D \subset \operatorname{dom} g$. Suppose $\bar{D}$ convex. Then the l.s.c. hull*

$$\overline{g + \delta(\cdot \mid D)} \quad \textit{of} \quad g + \delta(\cdot \mid D) \textit{ is equal to } g + \delta(\cdot \mid \bar{D}).$$

In particular $\inf_D g = \inf_{\bar{D}} g$.

Proof. Obviously $g + \delta(\cdot \mid \bar{D}) \leqslant \overline{g + \delta(\cdot \mid D)}$. Without loss of generality we may suppose that the affine subspace generated by D is $\mathbb{R}^d$ itself. So $\operatorname{int}(\operatorname{co} D) \neq \varnothing$. Let $x_0 \in \operatorname{int}(\operatorname{co} D)$, one has $x_0 \in \operatorname{int}(\operatorname{dom} g) \cap \bar{D}$.

(a) As g is continuous at x_0,

$$\begin{aligned}\overline{g + \delta(\cdot \mid D)}(x_0) &= \varliminf_{\substack{x \to x_0 \\ x \in D}} g(x) = g(x_0) \\ &= [g + \delta(\cdot \mid \bar{D})](x_0).\end{aligned}$$

(b) Let $x_1 \in \bar{D}$, $x_1 \neq x_0$, and prove $\overline{g + \delta(\cdot \mid D)}(x_1) \leqslant g(x_1)$. Let $x_\lambda = \lambda x_1 + (1 - \lambda) x_0$. When λ runs through $[0, 1[$, x_λ belongs to $\operatorname{int}(\operatorname{dom} g) \cap \bar{D}$, hence, by (a),

$$\overline{g + \delta(\cdot \mid D)}(x_\lambda) = g(x_\lambda).$$

On a one-dimensional interval like $[x_0, x_1]$, a convex function is u.s.c., so when it is l.s.c. it is continuous. Hence

$$\begin{aligned}\overline{g + \delta(\cdot \mid D)}(x_1) &= \varliminf_{\lambda \to 1^-} \overline{g + \delta(\cdot \mid D)}(x_\lambda) \\ &= \varliminf_{\lambda \to 1^-} g(x_\lambda) = g(x_1).\end{aligned}$$

The last formula is easy.

APPENDIX 2

LEMMA A2. *Let Q be a multifunction on a topological space Ω to the convex subsets of $\mathbb{R}^d$. Then Q is l.s.c. on Ω iff $(x, z') \mapsto \delta^*(z' \mid Q(x))$ is l.s.c. on $\Omega \times \mathbb{R}^d$.*

Proof. Let $\varphi(x, z') = \delta^*(z' \mid Q(x))$.

(1) Suppose Q is l.s.c. Let $(x_0, z_0') \in \Omega \times \mathbb{R}^d$ and $r \in \mathbb{R}$, $r < \varphi(x_0, z_0')$. The set $W = \{(z, z') \in (\mathbb{R}^d)^2 \mid z \cdot z' > r\}$ is open. There exists $z_0 \in Q(x_0)$ such that $(z_0, z_0') \in W$. There exists U an open neighborhood of z_0 and U' an open neighborhood of z_0' such that $U \times U'$ is contained in W. As Q is l.s.c. and $z_0 \in Q(x_0) \cap U$, there exists a neighborhood V of x_0 such that $\forall x \in V$, $Q(x) \cap U \neq \varnothing$. Hence

$$(x, z') \in V \times U' \Rightarrow \varphi(x, z') \geqslant z_x \cdot z' \qquad (\text{where } z_x \in Q(x) \cap U)$$
$$> r.$$

Thus φ is l.s.c. at (x_0, z_0').

(2) Suppose φ is l.s.c. and Q is not l.s.c. at x_0. Let U be an open subset of $\mathbb{R}^d$ such that $Q(x_0) \cap U \neq \varnothing$. We may suppose U convex and $0 \in Q(x_0) \cap U$. Thus $\varphi(x_0, \cdot) \geqslant 0$. There exists a generalized sequence (y_α) such that $y_\alpha \to x_0$ and $Q(y_\alpha) \cap U = \varnothing$. By the Hahn–Banach theorem $\exists z_\alpha'$ and $r \in \mathbb{R}$ such that

$$\varphi(y_\alpha, z_\alpha') \leqslant r \leqslant \inf_{z \in U} z \cdot z_\alpha'.$$

We may suppose $r = -1$. Thus $z_\alpha' \in \{z' \mid \forall z \in U,\ z \cdot z' \geqslant -1\}$, which is an equicontinuous set (here a bounded subset of $\mathbb{R}^d$). Let z' be a cluster point of the generalized sequence (z_α'). By the lower semi-continuity of φ, $\varphi(x_0, z') \leqslant -1$, which is a contradiction.

Remark. This improves in one direction II.21 of Castaing and Valadier [15].

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